HERIOT WATT UNIVERSITY

SCHOOL OF MATHEMATICAL AND COMPUTER SCIENCES

Department of Actuarial Mathematics & Statistics

F78AB

ACTUARIAL & FINANCIAL MATHEMATICS B

Semester 2 2021/22

Duration: 2 Hours

Total marks: 60

ANSWER **ALL** EIGHT QUESTIONS

A UNIVERSITY APPROVED CALCULATOR MAY BE USED

Question 1

You take a loan of \$50 from a bank to initiate a small project at time $t = 0$. The project provides returns of \$10 at time $t = 1$; \$20 at time $t = 2$; ...; \$100 at time $t = 10$. Immediately after the last payment at time $t = 10$ is received, the project is terminated.

(a) (i) Write down an expression for the net present value of the project at time $t = 0$.

(1 mark)

(ii) Show by substitution that the internal rate of return for the project is approximately 53.7% p.a.

(1 mark)

The interest charged by the bank for the loan is 5% per annum effective. Money can be invested to earn interest at 4% per annum effective.

(b) Find the profits earned at $t = 10$ if you decide to pay off the loan using one of the two options below.

(i) Option 1: Pay off the loan by level annual repayments.

(3 marks)

(ii) Option 2: Pay off the loan as quickly as possible. (Hint: Use trial-and-error to determine the time at which the loan is paid off.)

(4 marks)

(c) State which option in (b) would you choose and briefly explain the reason for your choice.

(1 mark)

(Total: 10 marks)

Question 2

Information about an investment fund is given in the following table.

Date	**Cashflows (in $)**	**Value of the fund (in $)**
1 January 2018		100
31 December 2018		110
1 January 2019	+50	
31 December 2019		170
1 January 2020	+50	
31 December 2020		200

Answer the following based on the table above.

(a) Calculate the Money-Weighted Rate of Return from 1 January 2018 to 31 December 2020.

(1 mark)

(b) Calculate the Linked Internal Rate of Return (LIRR) from 1 January 2018 to 31 December 2020, based on the three sub-periods: the calendar year of 2018, 2019 and 2020.

(3 marks)

(c) Explain which of the measures of the fund manager performance, calculated above, reflects their true performance.

(2 marks)

(Total: 6 marks)

Question 3

A bond, of nominal value \$1,000, is issued on 1 December 2019, with a term to redemption of 2 years and a coupon rate of \$50 payable annually, and the bond is redeemed at par. An investor purchases the entire issuance of the bond on 1 December 2019 at price \$960 and holds it until redemption.

(a) Find the money yield.

(2 marks)

(b) Calculate the real yield based on the following Retail Price Index (RPI).

Date	1 December 2019	1 December 2020	1 December 2021
RPI at the date	100	110	125

(3 marks)

(c) Explain why the money yield calculated in part (a) is not the same as the real yield calculated in part (b).

(1 mark)

(Total: 6 marks)

Question 4

Let the random variable i_t be the effective interest p.a. from time $t-1$ to time t, $t = 1, 2, \ldots, n$, with

$$i_t = \begin{cases} r_1 & \text{with probability of } \frac{1}{2} \\ r_2 & \text{with probability of } \frac{1}{2} \end{cases},$$

for constants $r_1, r_2 > -1$. Assume that $i_1, i_2, \ldots, i_n$ are independent random variables. Let X be the present value at time $t = 0$ of an investment of 1 unit made at integer time $t = n$.

(a) Write down the following expressions in term of n, r_1 and r_2.

(i) The expected value of X.

(3 marks)

(ii) The variance of X.

(4 marks)

(b) Write down an expression for the expected value in (a)(i) if the annual effective rate of interest is constant.

(1 mark)

(Total: 8 marks)

Question 5

(a) Consider the payment stream whose value at the start of the payment stream is represented by $\left(\overline{I\overline{a}}\right)_{\overline{10}|}$. Calculate the ratio of the total payments made under $\left(\overline{I\overline{a}}\right)_{\overline{10}|}$ during the second half of the term of the annuity to those made during the first half (i.e., the ratio of the undiscounted payments).

(2 marks)

(b) Evaluate $\left(\overline{I\overline{a}}\right)_{\overline{\infty}|}$ at a force of interest of 0.08 per annum.

(2 marks)

(c) Payments under a continuous perpetuity are made at the rate $\rho(t) = \left(1+k\right)^t$, $t \in [0, \infty)$. The annual effective rate of interest is i, where $0 < k < i$. Show from the first principal that the present value of the perpetuity is $\dfrac{1}{\ln(1+i) - \ln(1+k)}$.

(4 marks)

(d) Assume that today corresponds to time 0 and that time is measured in years. A cashflow is to be paid continuously, starting in one year's time and then being paid for 13 years. The rate of payment at time $t \in [1, 14]$ is $t^2 - 1$ per annum. The force of interest at time $t \geq 0$ is $(1+t)^{-1}$ per annum. Find the present value of the annuity.

(4 marks)

(Total: 12 marks)

Question 6

At time $t = 0$, an investor has a portfolio consisting of holdings in a 4-year and a 8-year zero-coupon bond. The investor has liabilities of \$1 million and \$2 million due in 4 years' and 5 years' time, respectively.
Calculate the amount needed to invest in the two bonds such that this portfolio is immune to small changes in the interest rate, given that the effective rate of interest is 5% per annum.

(Total: 7 marks)

Question 7

A 1-year forward contract is issued on 1 January 2022 on a dividend-paying stock that currently sells for \$100. It is known that dividends of \$5 per unit-holding of stock will be paid after 4 and 8 months. The force of interest is 3% per year.

(a) Assume an investor is in the long position of the contract. Use a replicating portfolio to calculate the no-arbitrage forward price for this investor.

(6 marks)

(b) Write an expression for the payoff at the delivery time of a short position in this forward contract.

(1 mark)

(Total: 7 marks)

Question 8

The prices of zero-coupon bonds (ZCB) currently available in the market are:

Time to maturity (years)	**ZCB price per $1 nominal**
1	0.95420
2	0.90703
3	0.85892

(a) Calculate the 2-year spot rate and express it as an annual effective rate.

(2 marks)

(b) Calculate the 1-year forward rate, applying from time 2 years, and express it as an annual effective rate.

(2 marks)

(Total: 4 marks)

END OF PAPER